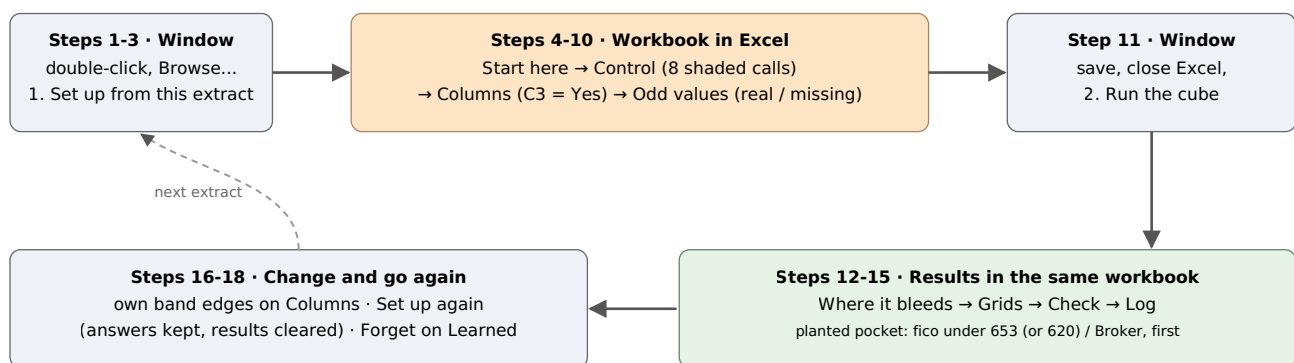


Origination analysis: the window and the workbook

Who this is for: the analyst preparing an origination analysis for a bank's line of business (LOB). No command line is needed. **What you finish with:** one workbook beside the loan extract. It holds your answers and a ranked list of the pockets of the book that lose more than their share, with a test on each. **Walked:** 25 September 2026, on commit f2f1032, with a synthetic book of 5,000 loans. The book hides one bad pocket, scores under 620 in the Broker channel. No real bank data was used.

Build note. Another session changed the product while this walk ran. Commit 8ea7b3d (14:14) added a "Cut by it?" column to the Columns tab and redrew the Grids tab, so on that build the Columns letters move one to the right from column D: band edges are in column F, not E. Steps 6, 10, 13 and 16 need a re-walk on the current build. Everything else matched when re-checked.



Peach = where you answer (Excel). Grey = the window. Green = where you read the results. Nothing is typed at a command line.

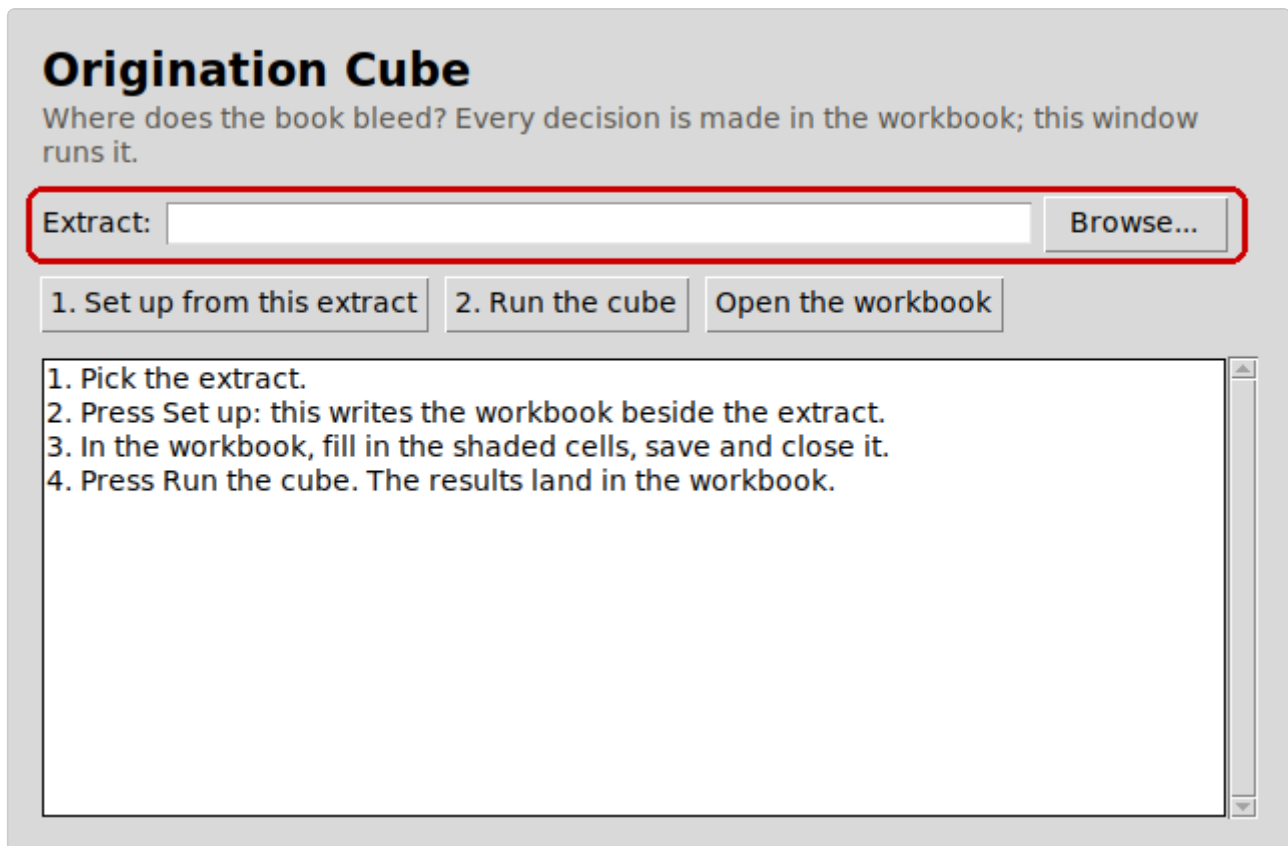
The route: two buttons in the window, one workbook in Excel.

Before you start

- **Python 3.10 or later**, and **Install add-ons.bat** run once. You only need to do this the first time.
- **Excel.**
- **The loan extract** (.csv or .xlsx) in its own folder. It needs a loan number, the booked amount, a yes/no outcome, GCO dollars and RANR dollars.
- **One folder per extract.** Don't copy last quarter's folder and drop a new extract into it under the same name. The workbook keeps reading the old extract (defect 1).

Step 1: Open the window

Double-click **Origination Cube.pyw** in the origination-cube folder.

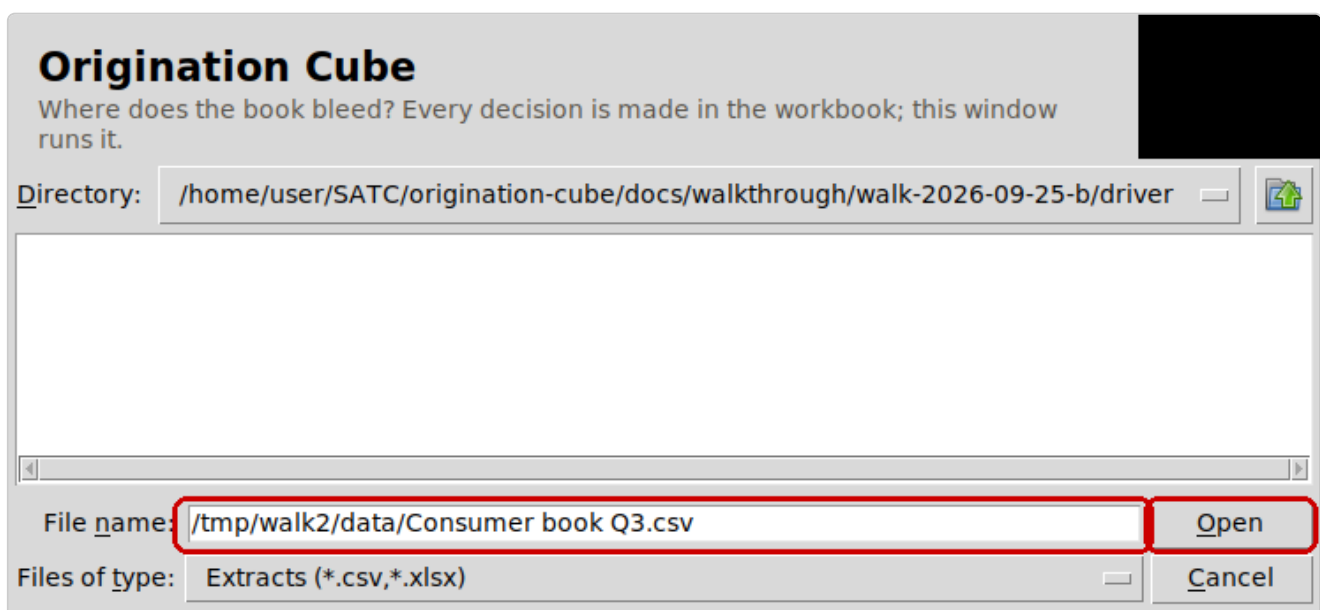


Right: a small window titled Origination Cube, with an empty Extract box and three buttons. The box underneath lists four steps.

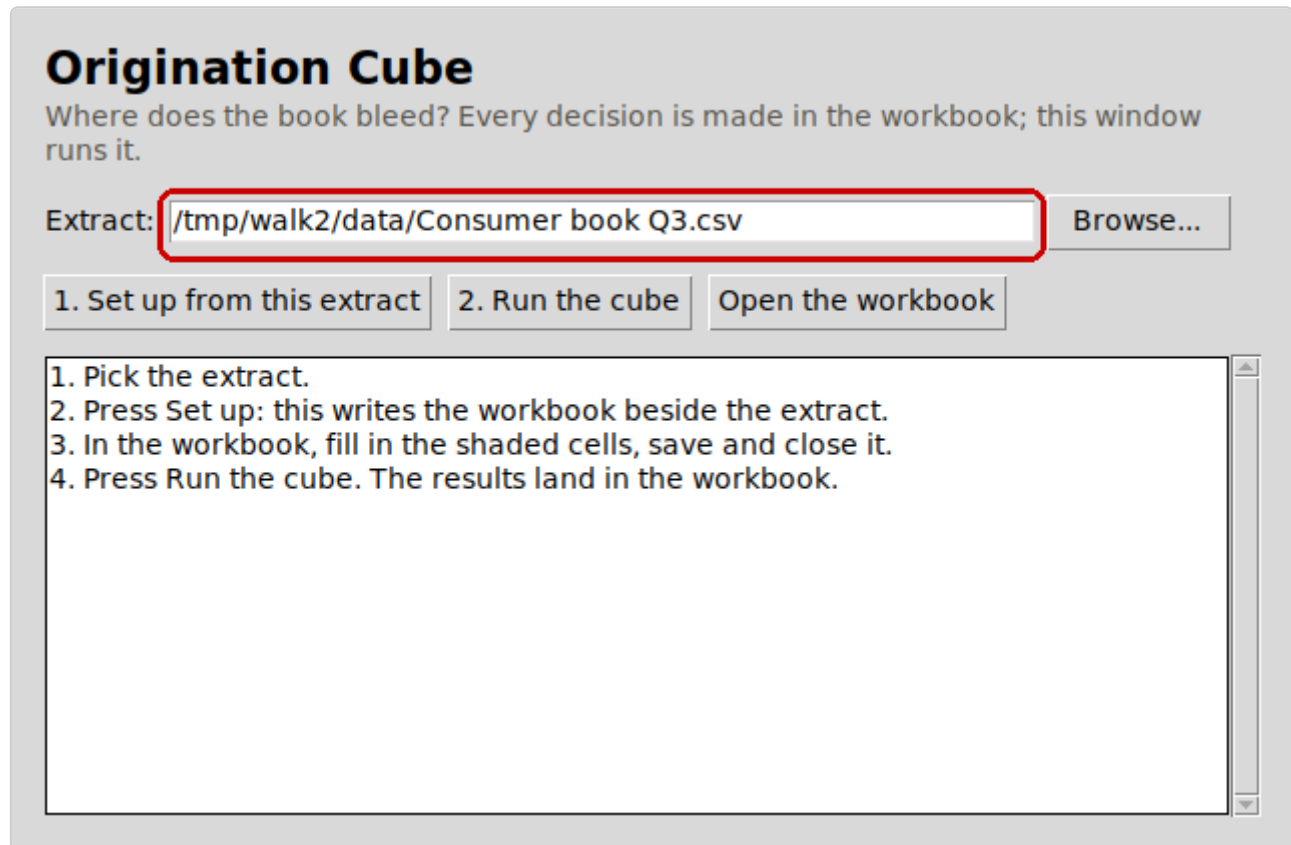
Wrong: if the box says something needs installing, run `Install add-ons.bat` once, then open the window again.

Step 2: Pick the extract

Press **Browse...** and pick the loan file.



On Windows this is the usual Open dialog. The picture shows the same step on the test machine.



Right: the full path of the extract is in the Extract box. The window remembers it for next time.

Step 3: Press 1. Set up from this extract

Origination Cube
Where does the book bleed? Every decision is made in the workbook; this window runs it.

Extract:

Set up Consumer book Q3 - Origination Cube.xlsx from Consumer book Q3.csv:
5,000 loans, 7 columns.
5 thing(s) to look at first, on the Columns tab.
Next: fill in the shaded cells on Control and Columns, save, close, and press Run.

Right: the box says the workbook was set up, with the number of loans and columns. Here that's 5,000 loans and 7 columns, and "5 thing(s) to look at first, on the Columns tab". A workbook called <extract name> - Origination Cube.xlsx now sits beside the extract.

Check the loan count against what the bank sent. If it's wrong, you picked the wrong file.

Step 4: Open the workbook: Start here

Press **Open the workbook**, or double-click it in its folder. It opens on **Start here**.

Origination Cube

Where does the book bleed? Set up from Consumer book Q3.csv: 5,000 loans, 7 columns.

Step	Where	What to do
1	Control	Fill in the shaded cells. Those are our calls: what's material, how many loans is enough, how much worse counts as worse.
2	Columns	Check what each column is; anything shaded has a reason beside it. Fix any that's wrong from the list, then set Checked every column to Yes.
3	Odd values	Answer what you can: real, or missing. Anything unanswered is used as it is.
4	Launcher	Save, close this workbook, and press Run the cube. Results land in the tabs on the right.

Where things stand

Calls still to make on Control	7
Things to look at first on Columns	5
Odd values found	2
Last run	not run yet

Step	Where	What to do
1	Control	Fill in the shaded cells. Those are our calls: what's material, how many loans is enough, how much worse counts as worse.
2	Columns	Check what each column is; anything shaded has a reason beside it. Fix any that's wrong from the list, then set Checked every column to Yes.
3	Odd values	Answer what you can: real, or missing. Anything unanswered is used as it is.
4	Launcher	Save, close this workbook, and press Run the cube. Results land in the tabs on the right.
Where things stand		
	Calls still to make on Control	7
	Things to look at first on Columns	5
	Odd values found	2
	Last run	not run yet

Right: four steps (Control, Columns, Odd values, Launcher) and a Where things stand block. On a new extract it says 8 calls to make on Control, 5 things to look at on Columns, 2 odd values, and not run yet.

Watch out: this block is only worked out at Set up. After you answer Control and run, it still says 8 (defect 9). Don't use it to tell whether you're done.

Step 5: Control: see what needs an answer

Click the **Control** tab.

Control center

Fill in the shaded cells before running. Those are the calls we make on every job: what's material, how many loans is enough to matter, and how much worse than its peers a pocket has to be. Everything else starts on a reasonable setting, so change it only if the population calls for it. Pick from the list, or type your own number in the next column and that wins. Change anything here and press Run again: the results are worked out when you run, not in the cells.

Setting	Choose	Or enter your own	In use	What it means
What the cube looks at				
How old a loan must be to count				Needs an answer before we run. Pick one, or enter your own.
How columns are recognised				
A number column with this many values or fewer is a category	12 (recommended)		2	Term (36, 48, 60) or a grade from 1 to 10 is read as categories, not cut into bands.
A text column with more values than this is too fine to cut by	50 (recommended)		0	State (50 or so) is still a dimension; a city or a dealer name is raised as a question instead.
How many bands a number column is cut into	5 (recommended)		5	Fine enough to see a trend, with each band holding a fifth of the loans.
Where the band edges fall	Each band holds about the same number of loans (recommended)	n/a	Each band holds about the same number of loans	Every band can be tested equally well. Edges land on odd numbers like 663.
Is it enough loans to matter				
Fewest loans in a pocket before it is tested				Needs an answer before we run. Pick one, or enter your own.
Fewest loans with a loss before a loss rate is tested				Needs an answer before we run. Pick one, or enter your own.
Is it material				
Smallest excess loss worth reporting				Needs an answer before we run. Pick one, or enter your own.
Does it underperform				
What a pocket is judged against		n/a		Needs an answer before we run. Pick one, or enter your own.
How much worse than its comparison a pocket must be				Needs an answer before we run. Pick one, or enter your own.
How much better than its comparison a pocket must be				Needs an answer before we run. Pick one, or enter your own.
Is it real				
How sure a difference must be before it counts				Needs an answer before we run. Pick one, or enter your own.
How often a real gap should be caught	80% (recommended)		80%	The usual standard. Used only to say how big a gap each pocket's size could show; it never flags anything.
Allowing for testing many pockets at once	Hold down the share of lucky finds (Benjamini-Hochberg) (recommended)	n/a	Hold down the share of lucky finds (Benjamini-Hochberg)	Of the pockets that pass, about 5% at 95% are expected to be luck. Suits scanning for leads.

Shaded = still needs an answer				
Setting	Choose	Or enter your own	In use	What it means
What the cube looks at				
How old a loan must be to count				Needs an answer before we run. Pick one, or enter your own.
How columns are recognised				
A number column with this many values or fewer is a category	12 (recommended)		12	Term (36, 48, 60) or a grade from 1 to 10 is read as categories, not cut into bands.
A text column with more values than this is too fine to cut by	50 (recommended)		50	State (50 or so) is still a dimension; a city or a dealer name is raised as a question instead.
How many bands a number column is cut into	5 (recommended)		5	Fine enough to see a trend, with each band holding a fifth of the loans.
Where the band edges fall	Each band holds about the same number of loans (recommended)	n/a	Each band holds about the same number of loans	Every band can be tested equally well. Edges land on odd numbers like 663.
Is it enough loans to matter				
Fewest loans in a pocket before it is tested				Needs an answer before we run. Pick one, or enter your own.
Fewest loans with a loss before a loss rate is tested				Needs an answer before we run. Pick one, or enter your own.
Is it material				
Smallest excess loss worth reporting				Needs an answer before we run. Pick one, or enter your own.
Does it underperform				
What a pocket is judged against		n/a		Needs an answer before we run. Pick one, or enter your own.
How much worse than its comparison a pocket must be				Needs an answer before we run. Pick one, or enter your own.
How much better than its comparison a pocket must be				Needs an answer before we run. Pick one, or enter your own.
Is it real				
How sure a difference must be before it counts				Needs an answer before we run. Pick one, or enter your own.
How often a real gap should be caught	80% (recommended)		80%	The usual standard. Used only to say how big a gap each pocket's size could show; it never flags anything.
Allowing for testing many pockets at once	Hold down the share of lucky finds (Benjamini-Hochberg) (recommended)	n/a	Hold down the share of lucky finds (Benjamini-Hochberg)	Of the pockets that pass, about 5% at 95% are expected to be luck. Suits scanning for leads.

Right: eight rows are shaded peach. They are the calls the tool will not make for you:

- how old a loan must be to count
- the fewest loans in a pocket before it's tested
- the fewest loans with a loss
- the smallest excess loss worth reporting
- what a pocket is judged against
- how much worse counts
- how much better counts
- how sure you need to be

The other rows start on a recommended setting, and grey n/a cells mean you can only pick from the list.

Step 6: Columns: see what each column was taken to be

Click the **Columns** tab.

Columns						
What each column is. Anything shaded in the Look first column is worth a second look before you run; the reason is beside it. Change a column's meaning from the list if it's wrong. Then set Checked every column to Yes; nothing runs until you do, and what you confirm is remembered for next time.						
Checked every column? ☐						
Column	What it is	Yes means (outcome only)	Band edges (optional)	Look first	Why this was suggested	Blank Samples
LOAN_NBR	key				name contains 'loan'; a different value on every row	0% L0000000, L0000001, L0000002
FICO	fico			100 rows look odd (repeated value). Used as recorded until you answer it at the bottom.	name contains 'fico'; 98% of values between 300 and 850	0% -9999, 680, 708
CHANNEL	category			suggested as category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.	3 different values	0% Online, Branch, Broker
ORIG_BAL	booked				name contains 'bal'; positive on every loan	0% 7655.75, 8189.94, 35307.6
BAD_FLAG	outcome			1 value(s) are neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	name contains 'bad'; 0 and 1 (4.6% are 1)	0% 0, 2, 1
GCO_AMT	gco			1 value(s) aren't numbers among 4,999 that are. They're left out of any rate using this column and counted.	name contains 'gco'; zero on 95% of loans, never negative	0% 0.0, #N/A, 2899.4
RANR_AMT	ranr			1,720 rows look odd (negatives). Used as recorded until you answer it at the bottom.	name contains 'ranr'; numbers	0% -143.52, -177.5, 139.27

Right: one row per column of the extract, each with a meaning in What it is and the reason beside it. Rows with something under **Look first** are shaded, and here there are five:

- **FICO:** 100 rows of -9999
- **CHANNEL:** suggested as a category from its shape alone
- **BAD_FLAG:** one value that is neither 0 nor 1
- **GCO_AMT:** one value that isn't a number
- **RANR_AMT:** 1,720 negative values

Checked every column? (C3) is shaded and empty.

Watch out: "answer it at the bottom" means the **Odd values** tab, not the bottom of this one (defect 10).

Step 7: Odd values, and Learned

Click **Odd values**.

Odd values				
Things in the data that could be a code rather than a real value. None of them stops a run: until you answer, the values are used as they are. Answer real, or missing (treated as blank and counted). Your answers are remembered.				
Column	What looks odd	Rows	Answer	Note
FICO	the value -9999 repeated far more than any other	100		
RANR_AMT	negative values in a column that is mostly positive	1,720		

Odd values				
Things in the data that could be a code rather than a real value. None of them stops a run: until you answer, the values are used as they are. Answer real, or missing (treated as blank and counted). Your answers are remembered.				
Column	What looks odd	Rows	Answer	Note
FICO	the value -9999 repeated far more than any other	100		
RANR_AMT	negative values in a column that is mostly positive	1,720		

Right: one row per value that might be a code. Here that's FICO's -9999 (100 rows) and RANR's negatives (1,720 rows), each with a shaded Answer cell.

The **Learned** tab lists what the tool has remembered from earlier runs. On the first extract it says Nothing learned yet.

Everything the cube has learned from files you confirmed. Set a row to Forget if it shouldn't have been learned; it's dropped the next time you press Run.

Keep?	Kind	Column	What it learned	First confirmed	Last confirmed	Times
			Nothing learned yet.			

Everything the cube has learned from files you confirmed. Set a row to Forget if it shouldn't have been learned; it's dropped the next time you press Run.

Keep?	Kind	Column	What it learned	First confirmed	Last confirmed	Times
			Nothing learned yet.			

Step 8: If you press Run too early

If you save, close and press **2. Run the cube** before answering, nothing runs.

Origination Cube

Where does the book bleed? Every decision is made in the workbook; this window runs it.

Extract: /tmp/walk2/data/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Control!C6: "How old a loan must be to count" needs an answer. Pick one, or enter your own in column D.
- Control!C13: "Fewest loans in a pocket before it is tested" needs an answer. Pick one, or enter your own in column D.
- Control!C14: "Fewest loans with a loss before a loss rate is tested" needs an answer. Pick one, or enter your own in column D.
- Control!C16: "Smallest excess loss worth reporting" needs an answer. Pick one, or enter your own in column D.
- Control!C18: "What a pocket is judged against" needs an answer. Pick one, or enter your own in column D.
- Control!C19: "How much worse than its comparison a pocket must be" needs an answer. Pick one, or enter your own in column D.

Right: "Couldn't run yet", then one line per thing to fix, each naming the tab and cell, e.g. Control!C6: "How old a loan must be to count" needs an answer. The same list is written to the **Log** tab.

Watch out:

- **Scroll the box.** The last line, Columns!C3: set Checked every column to Yes, is below the fold (defect 15).
- **Row 18 has no column D.** "What a pocket is judged against" only takes a pick from the list, whatever the message says (defect 11).

Step 9: Fill in the shaded cells on Control

Click each shaded **Choose** cell and pick from its list. To use a number that isn't offered, type it in **Or enter your own** instead, and your number wins. The walk used:

Setting	Answer
How old a loan must be to count	Every loan
Fewest loans in a pocket before it is tested	30
Fewest loans with a loss before a loss rate is tested	10
Smallest excess loss worth reporting	1% of the book's total losses
What a pocket is judged against	The rest of its band
How much worse	1.25 times
How much better	0.8 times
How sure	95%

Control center				
Fill in the shaded cells before running. Those are the calls we make on every job: what's material, how many loans is enough to matter, and how much worse than its peers a pocket has to be. Everything else starts on a reasonable setting, so change it only if the population calls for it. Pick from the list, or type your own number in the next column and that wins. Change anything here and press Run again: the results are worked out when you run, not in the cells.				
Shaded = still needs an answer				
Setting	Choose	Or enter your own	In use	What it means
What the cube looks at				
How old a loan must be to count	Every loan		Every loan	New loans have not had time to go bad, so a pocket full of recent loans looks better than it is.
How columns are recognised				
A number column with this many values or fewer is a category	12 (recommended)		12	Term (36, 48, 60) or a grade from 1 to 10 is read as categories, not cut into bands.
A text column with more values than this is too fine to cut by	50 (recommended)		50	State (50 or so) is still a dimension; a city or a dealer name is raised as a question instead.
How many bands a number column is cut into	5 (recommended)		5	Fine enough to see a trend, with each band holding a fifth of the loans.
Where the band edges fall	Each band holds about the same number of loans (recommended)	n/a	Each band holds about the same number of loans	Every band can be tested equally well. Edges land on odd numbers like 663.
Is it enough loans to matter				
Fewest loans in a pocket before it is tested		30	30	The old workbook's value. Stops a test on a handful of loans; the test itself decides whether a gap is luck.
Fewest loans with a loss before a loss rate is tested		10	10	A loss rate rests on how many loans lost money, not how many a pocket holds. Ten is a common floor.
Is it material				
Smallest excess loss worth reporting	1% of the book's total losses		1% of the book's total losses	Scales with the book, so the same setting works on any population.
Does it underperform				
What a pocket is judged against	The rest of its band	n/a	The rest of its band	Broker under 620 against the rest of under 620. Both comparisons are printed; this one decides the flag.
How much worse than its comparison a pocket must be	1.25 times		1.25 times	The old workbook's value. Moderate gaps; more pockets flagged.
How much better than its comparison a pocket must be	0.8 times		0.8 times	The old workbook's value.
Is it real				
How sure a difference must be before it counts	95%		95%	The usual standard. About 1 pocket in 20 with no real difference will still pass.
How often a real gap should be caught	80% (recommended)		80%	The usual standard. Used only to say how big a gap each pocket's size could show; it never flags anything.
Allowing for testing many pockets at once	Hold down the share of lucky finds (Benjamini-Hochberg) (recommended)	n/a	Hold down the share of lucky finds (Benjamini-Hochberg)	Of the pockets that pass, about 5% at 95% are expected to be luck. Suits scanning for leads.

Right: the peach shading is gone, and In use and What it means fill in for each row.

Watch out: if Excel turns your **95%** pick into the number 0.95, In use says **not an option** and the run refuses (defect 3). Until that's fixed, type **0.95** in Or enter your own on that row instead.

Step 10: Confirm the columns and answer the odd values

On **Columns**, check each What it is. Fix any that's wrong from its list, then set **Checked every column? (C3)** to **Yes**.

Columns

What each column is. Anything shaded in the Look first column is worth a second look before you run; the reason is inside it. Change a column's meaning from the list if it's wrong. Then set Checked every column to Yes: nothing runs until you do, and what you confirm is remembered for next time.

Checked every column? Yes

Column	What it is	Yes means (outcome only)	Band edges (optional)	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	key						
FICO	fico			100 rows look odd (repeated value). Used as recorded until you answer it at the bottom.	name contains 'loan'; a different value on every row	0% L0000000, L0000001, L0000002	
CHANNEL	category			suggested as category from its shape alone (3 different values). If it's something more specific, say so and it will be remembered.	name contains 'fico'; 98% of values between 300 and 850	0% -9999, 680, 708	
ORIG_BAL	looked				3 different values	0% Online, Branch, Broker	
BAD_FLAG	outcome			1 value(s) are neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	name contains 'bal'; positive on every loan	0% 7655.75, 8189.94, 35307.6	
GCO_AMT	gco			1 value(s) aren't numbers among 4,999 that are. They're left out of any rate using this column and counted.	name contains 'bad'; 0 and 1 (4.6% are 1)	0% 0, 2, 1	
RANR_AMT	ranr			1,720 rows look odd (negatives). Used as recorded until you answer it at the bottom.	name contains 'gco'; zero on 95% of loans, never negative	0% 0.0, #N/A, 2899.4	
					name contains 'ranr'; numbers	0% -143.52, -177.5, 139.27	

remembered for next time.

Checked every column? Yes

On **Odd values**, answer each row. The walk said FICO's -9999 is **missing** and RANR's negatives are **real**, because RANR is revenue and can be negative.

Odd values

Things in the data that could be a code rather than a real value. None of them stops a run: until you answer, the values are used as they are. Answer real, or missing (treated as blank and counted). Your answers are remembered.

Column	What looks odd	Rows	Answer	Note
FICO	the value -9999 repeated far more than any other	100	missing	
RANR_AMT	negative values in a column that is mostly positive	1,720	real	

Column	What looks odd	Rows	Answer	Note
FICO	the value -9999 repeated far more than any other		100 missing	
RANR_AMT	negative values in a column that is mostly positive		1,720 real	

Right: C3 reads Yes and is no longer shaded, and every Answer cell is filled. An unanswered odd value doesn't stop the run; its values are used as they are.

Step 11: Save, close, and press 2. Run the cube

Save the workbook and **close Excel**. Then press **2. Run the cube** in the window.

Origination Cube

Where does the book bleed? Every decision is made in the workbook; this window runs it.

Extract: /tmp/walk2/data/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 5,000 loans; 44 tie-out checks agree.
Worst for Outcome, share of loans: fico under 653 / channel Broker.
Worst for Outcome, share of booked dollars: fico under 653 / channel Broker.
Worst for GCO per booked dollar: fico under 653 / channel Broker.
Remembered 9 confirmed answers for next time.
Open Consumer book Q3 - Origination Cube.xlsx: the results are on Where it bleeds.

Right:

- "Ran on 5,000 loans; 44 tie-out checks agree."
- one Worst for line per measure
- "Remembered 9 confirmed answers"

On this book, the worst for the bad-loan share, the booked-weighted bad rate and GCO is **fico under 653 / channel Broker**. That's the planted pocket, seen through the recommended five equal bands.

Check the loan count. It should match step 3. If it doesn't, the workbook ran a different extract (defect 1).

Step 12: Read Where it bleeds

Open the workbook again. **Where it bleeds** lists every pocket that loses more than its share, largest first within each measure.

Where it bleeds

Every pocket losing more than its share (or, for RANR, earning less), largest first within each measure. The flag is judged against the rest of its band. Shaded red: worse. Shaded amber: worse, but could be luck.

Measure	Band	Pocket	Dimension	Pocket	Loans	Rate	Book rate	Excess	Material	vs rest of book	p	vs rest of band	p	Flag	Smallest gap it could show
Outcome, share of loans	fico	under 653	channel	Broker	330	15.76%	4.58%	37 yes		4.16x	0.0000	3.11x	0.0000 worse		1.77x
Outcome, share of loans	orig_bal	49,398 and over	channel	Broker	308	7.79%	4.58%	10 yes		1.78x	0.0809	2.45x	0.0409 worse		1.80x
Outcome, share of loans	orig_bal	under 16,126	channel	Broker	319	6.90%	4.58%	7 yes		1.56x	0.2037	1.38x	0.3194 worse, but could be luck		1.79x
Outcome, share of loans	orig_bal	16,126 to under 26,803	channel	Broker	352	6.25%	4.58%	6 yes		1.40x	0.3015	1.38x	0.0245 worse		1.75x
Outcome, share of loans	fico	747 and over	channel	Broker	315	6.35%	4.58%	6 yes		1.42x	0.3479	2.57x	0.0430 worse		1.79x
Outcome, share of loans	orig_bal	26,803 to under 38,549	channel	Broker	324	6.17%	4.58%	5 yes		1.34x	0.4484	1.26x	0.0661 worse, but could be luck		1.79x
Outcome, share of loans	fico	653 to under 685	channel	Broker	316	6.01%	4.58%	5 yes		1.34x	0.4484	1.26x	0.0661 worse, but could be luck		1.79x
Outcome, share of loans	orig_bal	under 653	channel	Branch	310	5.81%	4.58%	4 yes		1.29x	0.5030	0.57x	0.0450 better		1.80x
Outcome, share of loans	fico	under 16,126	channel	Branch	327	5.50%	4.58%	3 yes		1.22x	0.5138	0.97x	0.9226 in line		1.78x
Outcome, share of loans	orig_bal	38,549 to under 49,398	channel	Branch	309	5.18%	4.58%	2 below the line		1.14x	0.6835	0.94x	0.8935 in line		1.80x
Outcome, share of booked dollars	fico	under 653	channel	Broker	330	15.88%	4.60%	1,218,497 yes		4.18x	0.0000	2.84x	0.0807 worse		1.85x
Outcome, share of booked dollars	orig_bal	49,398 and over	channel	Broker	308	7.79%	4.60%	24,693 yes		1.78x	0.0809	2.45x	0.0409 worse		1.80x
Outcome, share of booked dollars	orig_bal	38,549 to under 49,398	channel	Broker	343	7.73%	4.60%	471,429 yes		1.80x	0.0744	1.84x	0.0656 worse, but could be luck		1.86x
Outcome, share of booked dollars	fico	747 and over	channel	Broker	315	6.99%	4.60%	249,809 yes		1.58x	0.2933	2.60x	0.0563 worse, but could be luck		1.90x
Outcome, share of booked dollars	fico	under 653	channel	Branch	310	6.45%	4.60%	181,009 yes		1.44x	0.4538	0.62x	0.1684 better, but could be luck		1.91x
Outcome, share of booked dollars	fico	653 to under 685	channel	Broker	316	6.29%	4.60%	169,998 yes		1.40x	0.4653	1.80x	0.3401 worse, but could be luck		1.90x
Outcome, share of booked dollars	orig_bal	26,803 to under 38,549	channel	Broker	324	5.99%	4.60%	148,724 yes		1.33x	0.5736	2.06x	0.0656 worse, but could be luck		1.89x
Outcome, share of booked dollars	orig_bal	under 16,126	channel	Broker	352	6.32%	4.60%	130,773 yes		1.40x	0.3247	3.25x	0.0319 worse		1.85x
Outcome, share of booked dollars	orig_bal	under 16,126	channel	Broker	319	6.71%	4.60%	72,725 below the line		1.47x	0.3221	1.39x	0.3327 worse, but could be luck		1.89x
Outcome, share of booked dollars	orig_bal	38,549 to under 49,398	channel	Branch	309	5.11%	4.60%	69,206 below the line		1.12x	0.6998	0.82x	0.8274 in line		1.91x
Outcome, share of booked dollars	fico	under 16,126	channel	Branch	327	5.58%	4.60%	34,304 below the line		1.22x	0.5314	1.04x	0.8869 in line		1.88x
Outcome, share of booked dollars	fico	under 653	channel	Online	321	4.81%	4.60%	23,003 below the line		1.05x	0.9883	0.42x	0.0098 better		1.89x
Outcome, share of booked dollars	fico	715 to under 747	channel	Branch	308	4.70%	4.60%	9,449 below the line		1.02x	0.9883	2.23x	0.1849 worse, but could be luck		1.91x
Outcome, share of booked dollars	fico	(missing by rule)	channel	Branch	39	5.39%	4.60%	7,743 below the line		1.17x	0.9883	1.24x	0.9130 too few loans to test		4.57x
Outcome, share of booked dollars	fico	(missing by rule)	channel	Broker	37	4.73%	4.60%	1,975 below the line		2.93x	0.9883	1.02x	0.9793 too few losses to test		4.01x
GCO per booked dollar	orig_bal	under 653	channel	Broker	330	8.70%	2.56%	662,485 yes		4.08x	0.0000	2.81x	0.0011 worse		1.91x
GCO per booked dollar	orig_bal	49,398 and over	channel	Broker	308	4.42%	2.56%	202,345 yes		1.86x	0.1099	2.49x	0.0541 worse		1.94x
GCO per booked dollar	orig_bal	38,549 to under 49,398	channel	Branch	343	3.99%	2.56%	215,235 yes		1.76x	0.1679	1.76x	0.1156 worse, but could be luck		1.89x
GCO per booked dollar	fico	under 653	channel	Branch	310	3.75%	2.56%	116,963 yes		1.51x	0.4406	0.67x	0.2669 better, but could be luck		1.94x
GCO per booked dollar	fico	653 to under 685	channel	Branch	316	3.56%	2.56%	106,446 yes		1.45x	0.4406	1.45x	0.4295 worse, but could be luck		1.93x
GCO per booked dollar	fico	747 and over	channel	Broker	315	3.58%	2.56%	106,194 yes		1.44x	0.4406	2.21x	0.1370 worse, but could be luck		1.93x
GCO per booked dollar	orig_bal	16,126 to under 26,803	channel	Broker	352	3.78%	2.56%	92,144 yes		1.51x	0.2898	3.20x	0.0341 worse		1.87x
GCO per booked dollar	orig_bal	26,803 to under 38,549	channel	Broker	324	3.26%	2.56%	74,620 yes		1.30x	0.4786	1.87x	0.1335 worse, but could be luck		1.91x
GCO per booked dollar	orig_bal	under 16,126	channel	Broker	319	3.98%	2.56%	48,500 yes		1.57x	0.2531	1.42x	0.3503 worse, but could be luck		1.92x
GCO per booked dollar	orig_bal	38,549 to under 49,398	channel	Branch	309	2.74%	2.56%	24,375 below the line		1.08x	0.8640	0.94x	0.9307 in line		1.94x
GCO per booked dollar	fico	715 to under 747	channel	Branch	308	2.76%	2.56%	10,028 below the line		1.08x	0.9293	2.41x	0.1911 worse, but could be luck		1.94x
GCO per booked dollar	orig_bal	under 16,126	channel	Branch	328	3.07%	2.56%	17,691 below the line		1.20x	0.6176	0.95x	0.9307 in line		1.91x
GCO per booked dollar	fico	(missing by rule)	channel	Broker	37	3.48%	2.56%	11,532 below the line		1.36x	0.9293	1.64x	0.9050 too few losses to test		4.19x
RANR per booked dollar	orig_bal	49,398 and over	channel	Online	379	0.18%	0.29%	23,047 yes		0.59x	0.0000	1.00x	0.9932 in line		1.28x
RANR per booked dollar	orig_bal	49,398 and over	channel	Branch	313	0.18%	0.29%	19,025 yes		0.59x	0.0000	1.00x	0.9932 in line		1.31x
RANR per booked dollar	orig_bal	49,398 and over	channel	Broker	308	0.18%	0.29%	18,773 yes		0.59x	0.0000	1.00x	0.9932 in line		1.32x
RANR per booked dollar	orig_bal	38,549 to under 49,398	channel	Online	348	0.20%	0.29%	11,566 yes		0.67x	0.0001	0.89x	0.9932 in line		1.30x
RANR per booked dollar	orig_bal	38,549 to under 49,398	channel	Broker	343	0.23%	0.29%	9,539 yes		0.77x	0.0041	1.07x	0.9932 in line		1.30x
RANR per booked dollar	orig_bal	38,549 to under 49,398	channel	Branch	309	0.23%	0.29%	8,821 yes		0.76x	0.0044	1.05x	0.9932 in line		1.32x
RANR per booked dollar	fico	under 653	channel	Broker	330	0.24%	0.29%	5,150 yes		0.83x	0.7271	0.80x	0.7483 worse, but could be luck		1.30x
RANR per booked dollar	fico	685 to under 715	channel	Branch	335	0.26%	0.29%	3,703 below the line		0.88x	0.7271	0.89x	0.7700 in line		1.30x
RANR per booked dollar	fico	747 and over	channel	Online	359	0.26%	0.29%	3,304 below the line		0.90x	0.7271	0.90x	0.7700 in line		1.29x
RANR per booked dollar	fico	under 653	channel	Online	321	0.27%	0.29%	2,303 below the line		0.92x	0.7271	0.92x	0.7990 in line		1.31x
RANR per booked dollar	fico	747 and over	channel	Branch	330	0.27%	0.29%	2,174 below the line		0.93x	0.7271	0.94x	0.8290 in line		1.30x
RANR per booked dollar	fico	685 to under 715	channel	Online	342	0.27%	0.29%	2,062 below the line		0.93x	0.7271	0.96x	0.8290 in line		1.30x
RANR per booked dollar	fico	653 to under 685	channel	Branch	313	0.28%	0.29%	1,620 below the line		0.94x	0.7271	0.89x	0.7700 in line		1.31x
RANR per booked dollar	fico	(missing by rule)	channel	Online	34	0.17%	0.29%	1,429 below the line		0.59x	0.7271	0.53x	0.7483 worse, but could be luck		2.03x
RANR per booked dollar	fico	(blank)	channel	Online	1	-2.17%	0.29%	201 below the line		-7.44x		0.53x	0.7483 worse, but could be luck		2.03x
RANR per booked dollar	fico	(missing by rule)	channel	Broker	37	0.29%	0.29%	4 below the line		1.00x	0.9973	1.12x	0.8290 in line		1.98x

Measure	Band	Pocket	Dimension	Pocket	Loans	Rate	Book rate	Excess	Material	vs rest of book	p	vs rest of band	p	Flag	Smallest gap it could show
Outcome, share of loans	fico	under 653	channel	Broker	330	15.76%	4.58%	37 yes		4.16x	0.0000	3.11x	0.0000 worse		1.77x
Outcome, share of loans	orig_bal	38,549 to under 49,398	channel	Broker	343	7.58%	4.58%	10 yes		1.74x	0.0809	1.78x	0.0793 worse, but could be luck		1.76x
Outcome, share of loans	orig_bal	49,398 and over	channel	Broker	308	7.79%	4.58%	10 yes		1.78x	0.0809	2.45x	0.0409 worse		1.80x
Outcome, share of loans	orig_bal	under 16,126	channel	Broker	319	6.90%	4.58%	7 yes		1.56x	0.2037	1.38x	0.3194 worse, but could be luck		1.79x
Outcome, share of loans	orig_bal	16,126 to under 26,803	channel	Broker	352	6.25%	4.58%	6 yes		1.40x	0.3015	1.38x	0.0245 worse		1.75x
Outcome, share of loans	fico	747 and over	channel	Broker	315	6.35%	4.58%	6 yes		1.42x	0.3479	2.57x	0.0430 worse		1.79x
Outcome, share of loans	orig_bal	26,803 to under 38,549	channel	Broker	324	6.17%	4.58%	5 yes		1.34x	0.3015	2.19x	0.0661 worse, but could be luck		1.78x
Outcome, share of loans	fico	653 to under 685	channel	Broker	316	6.01%	4.58%	5 yes		1.34x	0.4484	1.26x	0.0661 worse, but could be luck		1.79x
Outcome, share of loans	orig_bal	under 653	channel	Branch	310	5.81%	4.58%	4 yes		1.29x	0.5030	0.57x	0.0450 better		1.80x
Outcome, share of loans	orig_bal	under 16,126	channel	Branch	327	5.50%	4.58%	3 yes		1.22x	0.5138	0.97x	0.9226 in line		1.78x

Right: the first row of each loss measure is **fico under 653 / Broker**, ringed:

Measure	Rate	Book rate	vs rest of book	vs rest of band	Flag
Bad-loan share	15.76%	4.58%	4.16x	3.11x	worse
Booked-weighted bad rate	15.88%	4.60%	4.18x	2.84x	worse
GCO	8.70%	2.56%	4.08x	2.81x	worse

How to read a row:

- **Red rows** are worse, and the test says it isn't luck.
- **Amber rows** are worse, but could be luck.
- **Unshaded rows** have a flag of in line, better or too few ... to test.
- **The flag** is judged against whatever you chose in step 9 (here, the rest of its band). The title says which.
- **Material** says whether the excess clears the line you set. Below the line rows are still listed.

RANR reads the other way. RANR is revenue, so its rows are pockets earning less than their share. A multiple under 1 is the bad direction:

RANR per booked dollar	orig_bal	38,549 to under 49,398	channel	Online	348	0.20%	0.29%	11,566 yes	0.67x	0.0001	0.89x	0.9932 in line	1.30x
RANR per booked dollar	orig_bal	38,549 to under 49,398	channel	Broker	343	0.23%	0.29%	9,539 yes	0.77x	0.0041	1.07x	0.9932 in line	1.30x
RANR per booked dollar	orig_bal	38,549 to under 49,398	channel	Branch	309	0.23%	0.29%	8,821 yes	0.76x	0.0044	1.05x	0.9932 in line	1.32x
RANR per booked dollar	fico	under 653	channel	Broker	330	0.24%	0.29%	5,150 yes	0.83x	0.7271	0.80x	0.7483 worse, but could be luck	1.30x
RANR per booked dollar	fico	685 to under 715	channel	Branch	335	0.26%	0.29%	3,703 below the line	0.88x	0.7271	0.89x	0.7700 in line	1.30x
RANR per booked dollar	fico	747 and over	channel	Online	359	0.26%	0.29%	3,304 below the line	0.90x	0.7271	0.90x	0.7700 in line	1.29x
RANR per booked dollar	fico	under 653	channel	Online	321	0.27%	0.29%	2,303 below the line	0.92x	0.7271	0.92x	0.7990 in line	1.31x
RANR per booked dollar	fico	747 and over	channel	Branch	330	0.27%	0.29%	2,174 below the line	0.93x	0.7271	0.94x	0.8290 in line	1.30x
RANR per booked dollar	fico	685 to under 715	channel	Online	342	0.27%	0.29%	2,062 below the line	0.93x	0.7271	0.96x	0.8290 in line	1.30x
RANR per booked dollar	fico	653 to under 685	channel	Branch	313	0.28%	0.29%	1,620 below the line	0.94x	0.7271	0.89x	0.7700 in line	1.31x
RANR per booked dollar	fico	(missing by rule)	channel	Online	34	0.17%	0.29%	1,429 below the line	0.59x	0.7271	0.53x	0.7483 worse, but could be luck	2.03x
RANR per booked dollar	fico	(blank)	channel	Online	1	-2.17%	0.29%	201 below the line	-7.44x				
RANR per booked dollar	fico	(missing by rule)	channel	Broker	37	0.29%	0.29%	4 below the line	1.00x	0.9773	1.12x	0.8290 in line	1.98x
RANR per booked dollar	fico	685 to under 715	channel	Branch	335	0.26%	0.29%	3,703 below the line	0.88x	0.7271	0.89x	0.7700 in line	1.30x
RANR per booked dollar	fico	747 and over	channel	Online	359	0.26%	0.29%	3,304 below the line	0.90x	0.7271	0.90x	0.7700 in line	1.29x
RANR per booked dollar	fico	under 653	channel	Online	321	0.27%	0.29%	2,303 below the line	0.92x	0.7271	0.92x	0.7990 in line	1.31x
RANR per booked dollar	fico	747 and over	channel	Branch	330	0.27%	0.29%	2,174 below the line	0.93x	0.7271	0.94x	0.8290 in line	1.30x
RANR per booked dollar	fico	685 to under 715	channel	Online	342	0.27%	0.29%	2,062 below the line	0.93x	0.7271	0.96x	0.8290 in line	1.30x
RANR per booked dollar	fico	653 to under 685	channel	Branch	313	0.28%	0.29%	1,620 below the line	0.94x	0.7271	0.89x	0.7700 in line	1.31x
RANR per booked dollar	fico	(missing by rule)	channel	Online	34	0.17%	0.29%	1,429 below the line	0.59x	0.7271	0.53x	0.7483 worse, but could be luck	2.03x
RANR per booked dollar	fico	(blank)	channel	Online	1	-2.17%	0.29%	201 below the line	-7.44x				

Right: the RANR rows have vs rest of book below 1.00x. On the synthetic book the largest shortfalls are the biggest loans (49,398 and over), because the test data gives every loan a similar RANR. That's an artefact of the test data, not a finding.

Step 13: Grids

Grids shows each band crossed with each dimension, one block per measure. Each block has the rate first, then the rate over the book's rate.

Grids

Every band crossed with every dimension, one block per measure: each pocket's rate, then its rate over the book's.

fico x channel: Outcome, share of loans (COUNT(loans where BAD_FLAG = 1) / COUNT(loans))

Rate X channel: Outcome, share of loans (COUNT(loans where BAD_FLAG = 1) / COUNT(loans where BAD_FLAG = 0))					
Rate	Branch	Broker	Online	All	
under 653		5.81%	15.76%	4.36%	8.74%
653 to under 685		5.13%	6.01%	4.45%	5.18%
685 to under 715		1.49%	3.00%	2.92%	2.47%
715 to under 747		4.22%	3.81%	1.50%	3.13%
747 and over		3.33%	6.35%	1.67%	3.69%
(blank)				0.00%	0.00%
(missing by rule)		3.45%	2.70%	2.94%	3.00%
All		3.94%	6.93%	2.95%	4.58%
Over the book's rate					
under 653		1.27x	3.44x	0.95x	1.91x
653 to under 685		1.12x	1.31x	0.97x	1.13x
685 to under 715		0.32x	0.66x	0.64x	0.54x
715 to under 747		0.92x	0.83x	0.33x	0.68x
747 and over		0.73x	1.39x	0.36x	0.80x
(blank)				0.00x	0.00x
(missing by rule)		0.75x	0.59x	0.64x	0.65x
All		0.86x	1.51x	0.64x	1.00x

fico x channel: Outcome, share of booked dollars (SUM(ORIG_BAL where BAD_FLAG = 1) / SUM(ORIG_BAL))

Rate	Branch	Broker	Online	All	
under 653		6.45%	15.88%	4.81%	9.13%
653 to under 685		4.06%	6.29%	3.82%	4.69%
685 to under 715		1.07%	3.46%	2.54%	2.34%
715 to under 747		4.70%	3.11%	1.20%	2.92%
747 and over		3.71%	6.99%	1.82%	4.04%
(blank)				0.00%	0.00%
(missing by rule)		5.39%	4.73%	3.93%	4.64%
All		3.94%	7.14%	2.83%	4.60%
<i>Over the book's rate</i>					
under 653		1.40x	3.45x	1.05x	1.99x
653 to under 685		0.88x	1.37x	0.83x	1.02x
685 to under 715		0.23x	0.75x	0.55x	0.51x
715 to under 747		1.02x	0.68x	0.26x	0.63x
747 and over		0.81x	1.52x	0.40x	0.88x
(blank)				0.00x	0.00x
(missing by rule)		1.17x	1.03x	0.85x	1.01x
All		0.86x	1.55x	0.62x	1.00x

fico x channel: Outcome, share of loans (COUNT(loans where BAD_FLAG = 1) / COUNT(loans))

Rate	Branch	Broker	Online	All	
under 653		5.81%	15.76%	4.36%	8.74%
653 to under 685		5.13%	6.01%	4.45%	5.18%
685 to under 715		1.49%	3.00%	2.92%	2.47%
715 to under 747		4.22%	3.81%	1.50%	3.13%
747 and over		3.33%	6.35%	1.67%	3.69%
(blank)				0.00%	0.00%
(missing by rule)		3.45%	2.70%	2.94%	3.00%
All		3.94%	6.93%	2.95%	4.58%
<i>Over the book's rate</i>					
under 653		1.27x	3.44x	0.95x	1.91x
653 to under 685		1.12x	1.31x	0.97x	1.13x
685 to under 715		0.32x	0.66x	0.64x	0.54x
715 to under 747		0.92x	0.83x	0.33x	0.68x
747 and over		0.73x	1.39x	0.36x	0.80x
(blank)				0.00x	0.00x
(missing by rule)		0.75x	0.59x	0.64x	0.65x
All		0.86x	1.51x	0.64x	1.00x

Right: in fico x channel: Outcome, share of loans, the cell for under 653 / Broker is 15.76%, or 3.44x the book. It is the largest number in the block.

For RANR, **bigger is better**. A high multiple in the RANR block is good, not bad (the tab doesn't say so; defect 14):

All	0.88X	1.54X	0.61X	1.00X
fico x channel: RANR per booked dollar (SUM(RANR_AMT) / SUM(ORIG_BAL))				
Rate	Branch	Broker	Online	All
under 653	0.35%	0.24%	0.27%	0.28%
653 to under 685	0.28%	0.32%	0.30%	0.30%
685 to under 715	0.26%	0.31%	0.27%	0.28%
715 to under 747	0.33%	0.30%	0.30%	0.31%
747 and over	0.27%	0.32%	0.26%	0.28%
(blank)			-2.17%	-2.17%
(missing by rule)	0.37%	0.29%	0.17%	0.27%
All	0.30%	0.30%	0.28%	0.29%
<i>Over the book's rate</i>				
under 653	1.19x	0.84x	0.93x	0.98x
653 to under 685	0.95x	1.09x	1.04x	1.03x
685 to under 715	0.89x	1.06x	0.94x	0.96x
715 to under 747	1.15x	1.05x	1.05x	1.08x
747 and over	0.93x	1.09x	0.91x	0.97x
(blank)			-7.45x	-7.45x
(missing by rule)	1.26x	1.00x	0.59x	0.93x
All	1.02x	1.02x	0.96x	1.00x

fico x channel: RANR per booked dollar (SUM(RANR_AMT) / SUM(ORIG_BAL))					
Rate	Branch	Broker	Online	All	
under 653		0.35%	0.24%	0.27%	0.28%
653 to under 685		0.28%	0.32%	0.30%	0.30%
685 to under 715		0.26%	0.31%	0.27%	0.28%
715 to under 747		0.33%	0.30%	0.30%	0.31%
747 and over		0.27%	0.32%	0.26%	0.28%
(blank)				-2.17%	-2.17%
(missing by rule)		0.37%	0.29%	0.17%	0.27%
All		0.30%	0.30%	0.28%	0.29%
<i>Over the book's rate</i>					
under 653		1.19x	0.84x	0.93x	0.98x
653 to under 685		0.95x	1.09x	1.04x	1.03x
685 to under 715		0.89x	1.06x	0.94x	0.96x
715 to under 747		1.15x	1.05x	1.05x	1.08x
747 and over		0.93x	1.09x	0.91x	0.97x
(blank)				-7.45x	-7.45x
(missing by rule)		1.26x	1.00x	0.59x	0.93x
All		1.02x	1.02x	0.96x	1.00x

Step 14: Check

Check records what the run did:

- the tie-outs
- what was left out and why
- the band edges used
- what the book is big enough to show
- the materiality line per measure
- every setting in words

Check

What the run checked and used, for the record.

Loans run	5,000
Tie-out checks	44 of 44 agree: every grid adds up to the book
Left out of Outcome, share of loans	1 x BAD_FLAG flag not 0 or 1
Left out of Outcome, share of booked dollars	1 x ORIG_BAL blank; 1 x BAD_FLAG flag not 0 or 1
Left out of GCO per booked dollar	1 x GCO_AMT not a number; 1 x ORIG_BAL blank
Left out of RANR per booked dollar	1 x ORIG_BAL blank
Band edges used: fico	653, 685, 715, 747
Band edges used: orig_bal	16,126, 26,803, 38,549, 49,398
What this book can show	outcome_loans: about 2,806 loans before a 1.25x gap can be told from luck 80% of the time at 95% confidence (56.1% of this book's loans)
What this book can show	outcome_booked: about 3,530 loans before a 1.25x gap can be told from luck 80% of the time at 95% confidence (70.6% of this book's loans)
What this book can show	gco_rate: about 3,738 loans before a 1.25x gap can be told from luck 80% of the time at 95% confidence (74.8% of this book's loans)
What this book can show	ranr_rate: about 483 loans before a 1.25x gap can be told from luck 80% of the time at 95% confidence (9.7% of this book's loans)
Materiality line: outcome_loans	2
Materiality line: outcome_booked	75,288
Materiality line: gco_rate	41,976
Materiality line: ranr_rate	4,767
Setting: How old a loan must be to count	Every loan
Setting: Fewest loans in a pocket before it is	30
Setting: Fewest loans with a loss before a loss	10
Setting: Smallest excess loss worth reporting	1% of the book's total losses
Setting: What a pocket is judged against	The rest of its band
Setting: How much worse than its comparison	1.25 times
Setting: How much better than its comparison	0.8 times
Setting: How sure a difference must be before	95%
Setting: How often a real gap should be caught	80%
Setting: Allowing for testing many pockets at	Hold down the share of lucky finds (Benjamini-Hochberg)

Loans run	5,000
Tie-out checks	44 of 44 agree: every grid adds up to the book
Left out of Outcome, share of loans	1 x BAD_FLAG flag not 0 or 1
Left out of Outcome, share of booked dollars	1 x ORIG_BAL blank; 1 x BAD_FLAG flag not 0 or 1
Left out of GCO per booked dollar	1 x GCO_AMT not a number; 1 x ORIG_BAL blank
Left out of RANR per booked dollar	1 x ORIG_BAL blank
Band edges used: fico	653, 685, 715, 747
Band edges used: orig_bal	16,126, 26,803, 38,549, 49,398
What this book can show	outcome_loans: about 2,806 loans before a 1.25x gap can be told from luck 80% of the time at 95% confidence (56.1% of this book's loans)
What this book can show	outcome_booked: about 3,530 loans before a 1.25x gap can be told from luck 80% of the time at 95% confidence (70.6% of this book's loans)
What this book can show	gco_rate: about 3,738 loans before a 1.25x gap can be told from luck 80% of the time at 95% confidence (74.8% of this book's loans)
What this book can show	ranr_rate: about 483 loans before a 1.25x gap can be told from luck 80% of the time at 95% confidence (9.7% of this book's loans)
Materiality line: outcome_loans	2
Materiality line: outcome_booked	75,288
Materiality line: gco_rate	41,976
Materiality line: ranr_rate	4,767
Setting: How old a loan must be to count	Every loan
Setting: Fewest loans in a pocket before it is	30
Setting: Fewest loans with a loss before a loss	10
Setting: Smallest excess loss worth reporting	1% of the book's total losses
Setting: What a pocket is judged against	The rest of its band
Setting: How much worse than its comparison	1.25 times
Setting: How much better than its comparison	0.8 times
Setting: How sure a difference must be before	95%
Setting: How often a real gap should be caught	80%
Setting: Allowing for testing many pockets at	Hold down the share of lucky finds (Benjamini-Hochberg)

Right: "44 of 44 agree", and the settings match what you chose in step 9. The BAD_FLAG value of 2 and the GCO #N/A are counted as left out.

Step 15: Log, and Start here

Log keeps every run, newest first, including refused ones.

Ran on 5,000 loans; 44 tie-out checks agree.

Start here shows the time of the last run:

Origination Cube

Where does the book bleed? Set up from Consumer book Q3.csv: 5,000 loans, 7 columns.

Step	Where	What to do
1	Control	Fill in the shaded cells. Those are our calls: what's material, how many loans is enough, how much worse counts as worse.
2	Columns	Check what each column is; anything shaded has a reason beside it. Fix any that's wrong from the list, then set Checked every column to Yes.
3	Odd values	Answer what you can: real, or missing. Anything unanswered is used as it is.
4	Launcher	Save, close this workbook, and press Run the cube. Results land in the tabs on the right.

Where things stand

Calls still to make on Control	7
Things to look at first on Columns	5
Odd values found	2
Last run	2026-09-25 14:12

Where things stand

Calls still to make on Control	7
Things to look at first on Columns	5
Odd values found	2
Last run	2026-09-25 14:12

Step 16: Use the LOB's own cut points

If the LOB already cuts scores at, say, 620, 680 and 740, type them on **Columns** in the **Band edges** cell of that column's row. Put a comma **and a space** between numbers: 620, 680, 740. Save, close, and Run.

Origination Cube

Where does the book bleed? Every decision is made in the workbook; this window runs it.

Extract: /tmp/walk2/data/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

~~Ran on 5,000 loans; 44 tie-out checks agree.~~

Worst for Outcome, share of loans: fico under 620 / channel Broker.

Worst for Outcome, share of booked dollars: fico under 620 / channel Broker.

Worst for GCO per booked dollar: fico under 620 / channel Broker.

~~Remembered 9 confirmed answers for next time.~~

Open Consumer book Q3 - Origination Cube.xlsx: the results are on Where it bleeds.

Right: the worst pocket now reads **fico under 620 / channel Broker**. On Where it bleeds it's 34.4% against a book of 4.58%, 9.01x the rest of the book:

Where it bleeds

Every pocket losing more than its share (or, for BAD, earning less), largest first within each measure. The flag is judged against the rest of its band. Shaded red: worse. Shaded amber: worse, but could be luck.

Measure	Band	Pocket	Dimension	Pocket	Loans	Rate	Book rate	Excess	Material	vs rest of book	p	vs rest of band	p	Flag	Smallest gap it could show
Outcome, share of loans	fico	under 620	channel	Broker	125	34.40%	4.58%	37 yes		9.01x	0.0000	10.66x	0.0000 worse		2.32x
Outcome, share of loans	fico	620 to under 680	channel	Branch	439	7.06%	4.58%	11 yes		1.63x	0.0713	1.39x	0.3423 worse, but could be luck		1.66x
Outcome, share of loans	orig_bal	38,549 to under 49,398	channel	Broker	343	7.58%	4.58%	10 yes		1.74x	0.0809	1.78x	0.0793 worse, but could be luck		1.76x
Outcome, share of loans	orig_bal	49,398 and over	channel	Broker	308	7.79%	4.58%	10 yes		1.78x	0.0809	2.45x	0.0409 worse		1.80x
Outcome, share of loans	orig_bal	under 16,126	channel	Broker	319	6.90%	4.58%	7 yes		1.56x	0.2037	1.38x	0.3194 worse, but could be luck		1.79x
Outcome, share of loans	orig_bal	16,126 to under 26,803	channel	Broker	352	6.25%	4.58%	6 yes		1.40x	0.3015	3.38x	0.0245 worse		1.75x
Outcome, share of loans	orig_bal	26,803 to under 38,549	channel	Broker	324	6.17%	4.58%	5 yes		1.38x	0.3015	2.19x	0.0661 worse, but could be luck		1.78x
Outcome, share of loans	fico	620 to under 680	channel	Broker	461	5.64%	4.58%	5 yes		1.26x	0.4018	0.98x	0.9804 in line		1.65x
Outcome, share of loans	fico	740 and over	channel	Broker	376	5.85%	4.58%	5 yes		1.31x	0.3906	2.29x	0.0402 worse		1.72x
Outcome, share of loans	orig_bal	under 16,126	channel	Branch	327	5.50%	4.58%	3 yes		1.22x	0.5138	0.97x	0.9226 in line		1.78x
Outcome, share of loans	orig_bal	38,549 to under 49,398	channel	Branch	309	5.18%	4.58%	2 below the line		1.14x	0.6835	0.94x	0.8935 in line		1.80x
Outcome, share of booked dollars	fico	under 620	channel	Broker	125	34.13%	4.60%	1,256,393 yes		8.96x	0.0000	9.59x	0.0000 worse		2.51x
Outcome, share of booked dollars	orig_bal	49,398 and over	channel	Broker	308	7.82%	4.60%	540,972 yes		1.85x	0.0744	2.43x	0.0487 worse		1.91x
Outcome, share of booked dollars	orig_bal	38,549 to under 49,398	channel	Broker	343	7.73%	4.60%	471,429 yes		1.80x	0.0744	1.84x	0.0656 worse, but could be luck		1.86x
Outcome, share of booked dollars	fico	620 to under 680	channel	Branch	439	6.69%	4.60%	295,352 yes		1.52x	0.2298	1.35x	0.4465 worse, but could be luck		1.75x
Outcome, share of booked dollars	fico	740 and over	channel	Broker	376	6.29%	4.60%	211,652 yes		1.41x	0.3467	2.34x	0.0655 worse, but could be luck		1.82x
Outcome, share of booked dollars	orig_bal	26,803 to under 38,549	channel	Broker	324	5.99%	4.60%	148,724 yes		1.33x	0.3736	2.06x	0.0656 worse, but could be luck		1.89x
Outcome, share of booked dollars	fico	620 to under 680	channel	Broker	461	5.59%	4.60%	148,188 yes		1.24x	0.5106	1.02x	0.9795 in line		1.73x
Outcome, share of booked dollars	orig_bal	16,126 to under 26,803	channel	Broker	352	6.32%	4.60%	130,773 yes		1.40x	0.3247	3.25x	0.0329 worse		1.85x
Outcome, share of booked dollars	orig_bal	under 16,126	channel	Broker	319	6.71%	4.60%	72,275 below the line		1.47x	0.3211	1.39x	0.3317 worse, but could be luck		1.89x
Outcome, share of booked dollars	orig_bal	38,549 to under 49,398	channel	Branch	309	5.11%	4.60%	69,206 below the line		1.12x	0.6998	0.92x	0.8274 in line		1.91x
Outcome, share of booked dollars	orig_bal	under 16,126	channel	Branch	327	5.58%	4.60%	34,394 below the line		1.22x	0.5314	1.04x	0.8869 in line		1.88x
Outcome, share of booked dollars	fico	(missing by rule)	channel	Branch	28	5.39%	4.60%	7,742 below the line		1.17x	0.9883	1.24x	0.9795 too few loans to test		4.57x
Outcome, share of booked dollars	fico	(missing by rule)	channel	Broker	37	4.73%	4.60%	1,625 below the line		1.03x	0.9883	1.03x	0.9795 too few losses to test		4.08x

Measure	Band	Pocket	Dimension	Pocket	Loans	Rate	Book rate	Excess	Material	vs rest of book	p	vs rest of band	p	Flag	Smallest gap it could show
Outcome, share of loans	fico	under 620	channel	Broker	125	34.40%	4.58%	37 yes		9.01x	0.0000	10.66x	0.0000 worse		2.32x
Outcome, share of loans	fico	620 to under 680	channel	Branch	439	7.06%	4.58%	11 yes		1.63x	0.0713	1.39x	0.3423 worse, but could be luck		1.66x
Outcome, share of loans	orig_bal	38,549 to under 49,398	channel	Broker	343	7.58%	4.58%	10 yes		1.74x	0.0809	1.78x	0.0793 worse, but could be luck		1.76x
Outcome, share of loans	orig_bal	49,398 and over	channel	Broker	308	7.79%	4.58%	10 yes		1.78x	0.0809	2.45x	0.0409 worse		1.80x
Outcome, share of loans	orig_bal	under 16,126	channel	Broker	319	6.90%	4.58%	7 yes		1.56x	0.2037	1.38x	0.3194 worse, but could be luck		1.79x
Outcome, share of loans	orig_bal	16,126 to under 26,803	channel	Broker	352	6.25%	4.58%	6 yes		1.40x	0.3015	3.38x	0.0245 worse		1.75x

Watch out: typed without spaces, 620,680,740 is read by Excel as one number (620 million). The tool accepts that as a single edge, and every screen prints it as 620,680,740, so it looks right and isn't (defect 2, Wrong turn A2 below). On Check, confirm Band edges used shows the right number of edges.

Step 17: The next extract: press Set up again

Press **1. Set up from this extract** again, on the same extract or a refreshed one with the same name.

Origination Cube

Where does the book bleed? Every decision is made in the workbook; this window runs it.

Extract: /tmp/walk2/data/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Set up Consumer book Q3 - Origination Cube.xlsx from Consumer book Q3.csv: 5,000 loans, 7 columns.

2 thing(s) to look at first, on the Columns tab.

Next: fill in the shaded cells on Control and Columns, save, close, and press Run.

Right: your Control answers, your Columns answers and C3, and your Odd values answers are all kept. Columns now says **Remembered: you confirmed this as ...** for each column. Only two things are left to look at (BAD_FLAG's 2 and GCO's #N/A), which are in the data itself.

Control center

Fill in the shaded cells before running. Those are the calls we make on every job: what's material, how many loans is enough to matter, and how much worse than its peers a pocket has to be. Everything else starts on a reasonable setting, so change it only if the population calls for it. Pick from the list, or type your own number in the next column and that wins. Change anything here and press Run again: the results are worked out when you run, not in the cells.

Shaded = still needs an answer

Setting	Choose	Or enter your own	In use	What it means
What the cube looks at				
How old a loan must be to count	Every loan		very loan	New loans have not had time to go bad, so a pocket full of recent loans looks better than it is.
How columns are recognised				
A number column with this many values or fewer is a category	12 (recommended)		2	Term (36, 48, 60) or a grade from 1 to 10 is read as categories, not cut into bands.
A text column with more values than this is too fine to cut by	50 (recommended)		0	State (50 or so) is still a dimension; a city or a dealer name is raised as a question instead.
How many bands a number column is cut into	5 (recommended)			Fine enough to see a trend, with each band holding a fifth of the loans.
Where the band edges fall	Each band holds about the same number of loans (recommended)	n/a	each band holds about the same number of loans	Every band can be tested equally well. Edges land on odd numbers like 663.
Is it enough loans to matter				
Fewest loans in a pocket before it is tested		30	0	The old workbook's value. Stops a test on a handful of loans; the test itself decides whether a gap is luck.
Fewest loans with a loss before a loss rate is tested		10	0	A loss rate rests on how many loans lost money, not how many a pocket holds. Ten is a common floor.
Is it material				
Smallest excess loss worth reporting	1% of the book's total losses		% of the book's total losses	Scales with the book, so the same setting works on any population.
Does it underperform				
What a pocket is judged against	The rest of its band	n/a	the rest of its band	Broker under 620 against the rest of under 620. Both comparisons are printed; this one decides the flag.
How much worse than its comparison a pocket must be	1.25 times		.25 times	The old workbook's value. Moderate gaps; more pockets flagged.
How much better than its comparison a pocket must be	0.8 times		.8 times	The old workbook's value.
Is it real				
How sure a difference must be before it counts	95%		5%	The usual standard. About 1 pocket in 20 with no real difference will still pass.
How often a real gap should be caught	80% (recommended)		80%	The usual standard. Used only to say how big a gap each pocket's size could show; it never flags anything.
Allowing for testing many pockets at once	Hold down the share of lucky finds (Benjamini-Hochberg) (recommended)	n/a	Hold down the share of lucky finds (Benjamini-Hochberg)	Of the pockets that pass, about 5% at 95% are expected to be luck. Suits scanning for leads.

Columns

What each column is. Anything shaded in the Look first column is worth a second look before you run: the reason is inside it. Change a column's meaning from the list if it's wrong. Then set Checked every column? Yes, nothing runs until you do, and what you confirm is remembered for next time.

Checked every column? Yes

Column	What it is	Yes means (outcome only)	Band edges (optional)	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	key				Remembered: you confirmed this as loan or application number on 2026-09-25	0% L0000000, L0000001, L0000002	
FICO	fico				Remembered: you confirmed this as a FICO score on 2026-09-25	0% -9999, 680, 708	
CHANNEL	category				Remembered: you confirmed this as a category to cut by (channel) on 2026-09-25	0% Online, Branch, Broker	
ORIG_BAL	booked				Remembered: you confirmed this as booked or loan amount, which weights the reporting-level rates on 2026-09-25	0% 7655.75, 8189.94, 35307.6	
BAD_FLAG	outcome			1 value(s) are neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	Remembered: you confirmed this as the yes/no outcome: charged off, ever delinquent, decided by the system on 2026-09-25	0% 0, 2, 1	
GCO_AMT	gco			1 value(s) aren't numbers among 4,999 that are. They're left out of any rate using this column and counted.	Remembered: you confirmed this as gross charge-off dollars on 2026-09-25	0% 0.0, #N/A, 2899.4	
RANR_AMT	ranr				Remembered: you confirmed this as RANR dollars on 2026-09-25	0% -143.52, -177.5, 139.27	

Column	What it is	Yes means (outcome only)	Band edges (optional)	Look first	Why this was suggested	Blank	Samples
LOAN_NBR	key				Remembered: you confirmed this as loan or application number on 2026-09-25	0% L0000000, L0000001, L0000002	
FICO	fico				Remembered: you confirmed this as a FICO score on 2026-09-25	0% -9999, 680, 708	
CHANNEL	category				Remembered: you confirmed this as a category to cut by (channel) on 2026-09-25	0% Online, Branch, Broker	
ORIG_BAL	booked				Remembered: you confirmed this as booked or loan amount, which weights the reporting-level rates on 2026-09-25	0% 7655.75, 8189.94, 35307.6	
BAD_FLAG	outcome			1 value(s) are neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	Remembered: you confirmed this as the yes/no outcome: charged off, ever delinquent, decided by the system on 2026-09-25	0% 0, 2, 1	
GCO_AMT	gco			1 value(s) aren't numbers among 4,999 that are. They're left out of any rate using this column and counted.	Remembered: you confirmed this as gross charge-off dollars on 2026-09-25	0% 0.0, #N/A, 2899.4	

Watch out:

- **Set up clears the results.** It removes Where it bleeds, Grids, Check and Log, and Start here goes back to not run yet (defect 6). If you need the last results, save a copy of the workbook before pressing Set up.
- **Check new columns yourself.** If the new extract has columns the old one didn't, C3 still says Yes, and nothing marks the new rows. Look for them and check them before you run (defect 4).

Origination Cube

Where does the book bleed? Set up from Consumer book Q3.csv: 5,000 loans, 7 columns.

Step	Where	What to do
1	Control	Fill in the shaded cells. Those are our calls: what's material, how many loans is enough, how much worse counts as worse.
2	Columns	Check what each column is; anything shaded has a reason beside it. Fix any that's wrong from the list, then set Checked every column to Yes.
3	Odd values	Answer what you can: real, or missing. Anything unanswered is used as it is.
4	Launcher	Save, close this workbook, and press Run the cube. Results land in the tabs on the right.

Where things stand

Calls still to make on Control	0
Things to look at first on Columns	2
Odd values found	2
Last run	not run yet

Where things stand

Calls still to make on Control	0
Things to look at first on Columns	2
Odd values found	2
Last run	not run yet

Step 18: Prune something learned

On **Learned**, set a row's **Keep?** to **Forget**, then save, close and Run.

Everything the cube has learned from files you confirmed. Set a row to Forget if it shouldn't have been learned; it's dropped the next time you press Run.

Keep?	Kind	Column	What it learned	First confirmed	Last confirmed	Times
Keep	column	BAD_FLAG	outcome	2026-09-25	2026-09-25	1
Forget	column	CHANNEL	category	2026-09-25	2026-09-25	1
Keep	column	FICO	fico	2026-09-25	2026-09-25	1
Keep	column	GCO_AMT	gco	2026-09-25	2026-09-25	1
Keep	column	LOAN_NBR	key	2026-09-25	2026-09-25	1
Keep	column	ORIG_BAL	booked	2026-09-25	2026-09-25	1
Keep	column	RANR_AMT	ranr	2026-09-25	2026-09-25	1
Keep	answer	FICO	-9999 means missing	2026-09-25	2026-09-25	1
Keep	answer	RANR_AMT	negative values are real	2026-09-25	2026-09-25	1

Everything the cube has learned from files you confirmed. Set a row to Forget if it shouldn't have been learned; it's dropped the next time you press Run.

Keep?	Kind	Column	What it learned	First confirmed	Last confirmed	Times
Keep	column	BAD_FLAG	outcome	2026-09-25	2026-09-25	1
Forget	column	CHANNEL	category	2026-09-25	2026-09-25	1
Keep	column	FICO	fico	2026-09-25	2026-09-25	1
Keep	column	GCO_AMT	gco	2026-09-25	2026-09-25	1
Keep	column	LOAN_NBR	key	2026-09-25	2026-09-25	1
Keep	column	ORIG_BAL	booked	2026-09-25	2026-09-25	1
Keep	column	RANR_AMT	ranr	2026-09-25	2026-09-25	1
Keep	answer	FICO	-9999 means missing	2026-09-25	2026-09-25	1
Keep	answer	RANR_AMT	negative values are real	2026-09-25	2026-09-25	1

What happened on this walk: the run finished, but CHANNEL came straight back as learned. It shows Keep, with Times reset to 1:

Origination Cube

Where does the book bleed? Every decision is made in the workbook; this window runs it.

Extract: /tmp/walk2/data/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 5,000 loans; 44 tie-out checks agree.
Worst for Outcome, share of loans: fico under 653 / channel Broker.
Worst for Outcome, share of booked dollars: fico under 653 / channel Broker.
Worst for GCO per booked dollar: fico under 653 / channel Broker.
Remembered 9 confirmed answers for next time.
Open Consumer book Q3 - Origination Cube.xlsx: the results are on Where it bleeds.

Everything the cube has learned from files you confirmed. Set a row to Forget if it shouldn't have been learned; it's dropped the next time you press Run.

Keep?	Kind	Column	What it learned	First confirmed	Last confirmed	Times
Keep	column	BAD_FLAG	outcome	2026-09-25	2026-09-25	2
Keep	column	CHANNEL	category	2026-09-25	2026-09-25	1
Keep	column	FICO	fico	2026-09-25	2026-09-25	2
Keep	column	GCO_AMT	gco	2026-09-25	2026-09-25	2
Keep	column	LOAN_NBR	key	2026-09-25	2026-09-25	2
Keep	column	ORIG_BAL	booked	2026-09-25	2026-09-25	2
Keep	column	RANR_AMT	ranr	2026-09-25	2026-09-25	2
Keep	answer	FICO	-9999 means missing	2026-09-25	2026-09-25	2
Keep	answer	RANR_AMT	negative values are real	2026-09-25	2026-09-25	2

Everything the cube has learned from files you confirmed. Set a row to Forget if it shouldn't have been learned; it's dropped the next time you press Run.

Keep?	Kind	Column	What it learned	First confirmed	Last confirmed	Times
Keep	column	BAD_FLAG	outcome	2026-09-25	2026-09-25	2
Keep	column	CHANNEL	category	2026-09-25	2026-09-25	1
Keep	column	FICO	fico	2026-09-25	2026-09-25	2
Keep	column	GCO_AMT	gco	2026-09-25	2026-09-25	2
Keep	column	LOAN_NBR	key	2026-09-25	2026-09-25	2
Keep	column	ORIG_BAL	booked	2026-09-25	2026-09-25	2
Keep	column	RANR_AMT	ranr	2026-09-25	2026-09-25	2
Keep	answer	FICO	-9999 means missing	2026-09-25	2026-09-25	2
Keep	answer	RANR_AMT	negative values are real	2026-09-25	2026-09-25	2

Watch out: Forget only sticks for a column that isn't in the workbook you ran (defect 5). To correct a wrong meaning, change it on the **Columns** tab instead. The new meaning replaces the old one when you run.

When something goes wrong

What you did	What the window says	What to do
Band edges not rising (620, 740, 680)	"Columns!E7: band edges for FICO must be rising numbers separated by commas, like	Fix the order in that cell.

What you did	What the window says	What to do
	620, 680, 740." (The window shows FICO in backticks.)	
Band edges typed without spaces (620,680,740)	Nothing. It runs, with one edge at 620,680,740.	Retype with spaces. Check Band edges used on Check.
An out-of-range number typed on Control (0.9 for "worse", 95 for "how sure")	One line per cell, e.g. Control!D22: ... needs a share between 0.5 and 0.999 ... got 95. Excel also refuses it as you type, unless it was pasted.	Type 0.95, not 95.
The only category column marked servicing	"Nothing is left to cut across: mark at least one column as a category (or term) on the Columns tab."	Put that column back to category.
Workbook still open in Excel	"... is open in Excel. Close it, then press Run again." (Set up says the same.)	Close Excel and press again.
Workbook copied to a new folder with a new extract of the same name	Nothing. It runs the old folder's extract.	Don't copy workbooks between folders. Set up fresh from the new extract.

Wrong turn A: band edges not rising

Origination Cube

Where does the book bleed? Every decision is made in the workbook; this window runs it.

Extract:

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Columns!E7: band edges for `FICO` must be rising numbers separated by commas, like 620, 680, 740

Wrong turn A2: band edges typed without spaces. The run goes through. The grid has one band, "under 620,680,740", and Check prints the edge the way you typed it:

Origination Cube

Where does the book bleed? Every decision is made in the workbook; this window runs it.

Extract: /tmp/walk2/data/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 5,000 loans; 44 tie-out checks agree.

Worst for Outcome, share of loans: fico under 620,680,740 / channel Broker.
Worst for Outcome, share of booked dollars: fico under 620,680,740 / channel Broker.

Worst for GCO per booked dollar: fico under 620,680,740 / channel Broker.

Remembered 9 confirmed answers for next time.

Open Consumer book Q3 - Origination Cube.xlsx: the results are on Where it bleeds.

Grids

Every band crossed with every dimension, one block per measure: each pocket's rate, then its rate over the book's.

fico x channel: Outcome, share of loans (COUNT(loans where BAD_FLAG = 1) / COUNT(loans))

Rate	Branch	Broker	Online	All
under 620,680,740	3.95%	7.02%	2.95%	4.61%
(blank)			0.00%	0.00%
(missing by rule)	3.45%	2.70%	2.94%	3.00%
All	3.94%	6.93%	2.95%	4.58%
<i>Over the book's rate</i>				
under 620,680,740	0.86x	1.53x	0.64x	1.01x
(blank)			0.00x	0.00x
(missing by rule)	0.75x	0.59x	0.64x	0.65x
All	0.86x	1.51x	0.64x	1.00x

fico x channel: Outcome, share of booked dollars (SUM(ORIG_BAL where BAD_FLAG = 1) / SUM(ORIG_BAL))

Rate	Branch	Broker	Online	All
under 620,680,740	3.91%	7.20%	2.80%	4.60%
(blank)			0.00%	0.00%
(missing by rule)	5.39%	4.73%	3.93%	4.64%
All	3.94%	7.14%	2.83%	4.60%
<i>Over the book's rate</i>				
under 620,680,740	0.85x	1.57x	0.61x	1.00x
(blank)			0.00x	0.00x
(missing by rule)	1.17x	1.03x	0.85x	1.01x
All	0.86x	1.55x	0.62x	1.00x

Count of GCO per booked dollar: GCO/BAL (COUNT(GCO) / SUM(ORIG_BAL))

fico x channel: Outcome, share of loans (COUNT(loans where BAD_FLAG = 1) / COUNT(loans))

Rate	Branch	Broker	Online	All
under 620,680,740	3.95%	7.02%	2.95%	4.61%
(blank)			0.00%	0.00%
(missing by rule)	3.45%	2.70%	2.94%	3.00%
All	3.94%	6.93%	2.95%	4.58%
<i>Over the book's rate</i>				
under 620,680,740	0.86x	1.53x	0.64x	1.01x
(blank)			0.00x	0.00x
(missing by rule)	0.75x	0.59x	0.64x	0.65x
All	0.86x	1.51x	0.64x	1.00x

Check

What the run checked and used, for the record.

Loans run	5,000
Tie-out checks	44 of 44 agree: every grid adds up to the book
Left out of Outcome, share of loans	1 x BAD_FLAG flag not 0 or 1
Left out of Outcome, share of booked dollars	1 x ORIG_BAL blank; 1 x BAD_FLAG flag not 0 or 1
Left out of GCO per booked dollar	1 x GCO_AMT not a number; 1 x ORIG_BAL blank
Left out of RANR per booked dollar	1 x ORIG_BAL blank
Band edges used: fico	620,680,740
Band edges used: orig_bal	16,126, 26,803, 38,549, 49,398
What this book can show	outcome_loans: about 2,506 loans before a 1.25x gap can be told from luck 80% of the time at 95% confidence (56.1% of this book's loans)
What this book can show	outcome_booked: about 3,530 loans before a 1.25x gap can be told from luck 80% of the time at 95% confidence (70.6% of this book's loans)
What this book can show	gco_rate: about 3,738 loans before a 1.25x gap can be told from luck 80% of the time at 95% confidence (74.8% of this book's loans)
What this book can show	ranr_rate: about 483 loans before a 1.25x gap can be told from luck 80% of the time at 95% confidence (9.7% of this book's loans)
Materiality line: outcome_loans	2
Materiality line: outcome_booked	75,288
Materiality line: gco_rate	41,976
Materiality line: ranr_rate	4,767
Setting: How old a loan must be to count	Every loan
Setting: Fewest loans in a pocket before it is	30
Setting: Fewest loans with a loss before a loss	10
Setting: Smallest excess loss worth reporting	1% of the book's total losses
Setting: What a pocket is judged against	The rest of its band
Setting: How much worse than its comparison	1.25 times
Setting: How much better than its comparison	0.8 times
Setting: How sure a difference must be before	95%
Setting: How often a real gap should be caught	80%
Setting: Allowing for testing many pockets at	Hold down the share of lucky finds (Benjamini-Hochberg)

Loans run	5,000
Tie-out checks	44 of 44 agree: every grid adds up to the book
Left out of Outcome, share of loans	1 x BAD_FLAG flag not 0 or 1
Left out of Outcome, share of booked dollars	1 x ORIG_BAL blank; 1 x BAD_FLAG flag not 0 or 1
Left out of GCO per booked dollar	1 x GCO_AMT not a number; 1 x ORIG_BAL blank
Left out of RANR per booked dollar	1 x ORIG_BAL blank
Band edges used: fico	620,680,740
Band edges used: orig_bal	16,126, 26,803, 38,549, 49,398
What this book can show	outcome_loans: about 2,506 loans before a 1.25x gap can be told from luck 80% of the time at 95% confidence (56.1% of this book's loans)

Wrong turn B: out-of-range numbers. The Control tab itself shows no problem (the values were pasted, which gets past Excel's check). The window names both cells:

Origination Cube

Where does the book bleed? Every decision is made in the workbook; this window runs it.

Extract: /tmp/walk2/data/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:

- Control!D19: "How much worse than its comparison a pocket must be" needs a multiple above 1, such as 1.4, from 1.01 to 100; got 0.9.
- Control!D22: "How sure a difference must be before it counts" needs a share between 0.5 and 0.999, from 0.5 to 0.999; got 95.

Control center

Fill in the shaded cells before running. Those are the calls we make on every job: what's material, how many loans is enough to matter, and how much worse than its peers a pocket has to be. Everything else starts on a reasonable setting, so change it only if the population calls for it. Pick from the list, or type your own number in the next column and that wins. Change anything here and press Run again: the results are worked out when you run, not in the cells.

Shaded = still needs an answer

Setting	Choose	Or enter your own	In use	What it means
What the cube looks at				
How old a loan must be to count	Every loan		Every loan	New loans have not had time to go bad, so a pocket full of recent loans looks better than it is.
How columns are recognised				
A number column with this many values or fewer is a category	12 (recommended)		12	Term (36, 48, 60) or a grade from 1 to 10 is read as categories, not cut into bands.
A text column with more values than this is too fine to cut by	50 (recommended)		50	State (50 or so) is still a dimension; a city or a dealer name is raised as a question instead.
How many bands a number column is cut into	5 (recommended)		5	Fine enough to see a trend, with each band holding a fifth of the loans.
Where the band edges fall	Each band holds about the same number of loans (recommended)	n/a	Each band holds about the same number of loans	Every band can be tested equally well. Edges land on odd numbers like 663.
Is it enough loans to matter				
Fewest loans in a pocket before it is tested		30	30	The old workbook's value. Stops a test on a handful of loans; the test itself decides whether a gap is luck.
Fewest loans with a loss before a loss rate is tested		10	10	A loss rate rests on how many loans lost money, not how many a pocket holds. Ten is a common floor.
Is it material				
Smallest excess loss worth reporting	1% of the book's total losses		1% of the book's total losses	Scales with the book, so the same setting works on any population.
Does it underperform				
What a pocket is judged against	The rest of its band	n/a	The rest of its band	Broker under 620 against the rest of under 620. Both comparisons are printed; this one decides the flag.
How much worse than its comparison a pocket must be	1.25 times	0.9	0.9	Your own value, in place of the options (a multiple above 1, such as 1.4).
How much better than its comparison a pocket must be	0.8 times		0.8 times	The old workbook's value.
Is it real				
How sure a difference must be before it counts	95%	95	95	Your own value, in place of the options (a share between 0.5 and 0.999).
How often a real gap should be caught	80% (recommended)		80%	The usual standard. Used only to say how big a gap each pocket's size could show; it never flags anything.
Allowing for testing many pockets at once	Hold down the share of lucky finds (Benjamin-Hochberg) (recommended)	n/a	Hold down the share of lucky finds (Benjamin-Hochberg)	Of the pockets that pass, about 5% at 95% are expected to be luck. Suits scanning for leads.

What a pocket is judged against	The rest of its band	n/a	The rest of its band	Broker under 620 against the rest of under 620. Both comparisons are printed; this one decides the flag.
How much worse than its comparison a pocket must be	1.25 times		0.9	0.9 Your own value, in place of the options (a multiple above 1, such as 1.4).
How much better than its comparison a pocket must be	0.8 times		0.8 times	The old workbook's value.
Is it real				
How sure a difference must be before it counts	95%		95	95 Your own value, in place of the options (a share between 0.5 and 0.999).
How often a real gap should be caught	80% (recommended)		80%	The usual standard. Used only to say how big a gap each pocket's size could show; it never flags anything.

Wrong turn C: the only category marked servicing

Origination Cube

Where does the book bleed? Every decision is made in the workbook; this window runs it.

Extract: /tmp/walk2/data/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Couldn't run yet. Fix these in the workbook, save, close, and press Run again:
- Nothing is left to cut across: mark at least one column as a category (or term) on the Columns tab

Wrong turn D: workbook open in Excel. This was simulated by locking the file, because Excel isn't available here. Run and Set up both refuse in words:

Origination Cube

Where does the book bleed? Every decision is made in the workbook; this window runs it.

Extract:

Consumer book Q3 - Origination Cube.xlsx is open in Excel. Close it, then press Run again.

Origination Cube

Where does the book bleed? Every decision is made in the workbook; this window runs it.

Extract:

Consumer book Q3 - Origination Cube.xlsx is open in Excel. Close it, then press Set up again.

Wrong turn E: a refreshed extract with two new columns. CURR_STATUS and DTI were added. Set up kept C3 = Yes, and nothing shades the new rows:

Columns						
What each column is. Anything shaded in the Look first column is worth a second look before you run; the reason is beside it. Change a column's meaning from the list if it's wrong. Then set Checked every column to Yes; nothing runs until you do, and what you confirm is remembered for next time.						
Checked every column? Yes						
Column	What it is	Yes means (outcome only)	Band edges (optional)	Look first	Why this was suggested	Blank Samples
LOAN_NBR	key				Remembered: you confirmed this as loan or application number on 2026-09-25	0% L0000000, L0000001, L0000002
FICO	fico		620, 680, 740		Remembered: you confirmed this as a FICO score on 2026-09-25	0% -9999, 680, 708
CHANNEL	category				Remembered: you confirmed this as a category to cut by (channel on 2026-09-25)	0% Online, Branch, Broker
ORIG_BAL	booked				Remembered: you confirmed this as booked or loan amount, which weights the reporting-level rates on 2026-09-25	0% 7655.75, 8189.94, 35307.6
BAD_FLAG	outcome			1 value(s) are neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	Remembered: you confirmed this as the yes/no outcome: charged off, ever delinquent, decided by the system on 2026-09-25	0% 0, 2, 1
GCO_AMT	gco			1 value(s) aren't numbers among 4,999 that are. They're left out of any rate using this column and counted.	Remembered: you confirmed this as gross charge-off dollars on 2026-09-25	0% 0.0, #N/A, 2899.4
RANR_AMT	ranr				Remembered: you confirmed this as RANR dollars on 2026-09-25	0% -143.52, -177.5, 139.27
CURR_STATUS	servicing				name contains 'status';	0% 60DPD, ChargedOff, Current
DTI	dti				name contains 'dti'; a ratio with median 30%	0% 0.479, 0.360, 0.429

Checked every column? Yes						
Column	What it is	Yes means (outcome only)	Band edges (optional)	Look first	Why this was suggested	Blank Samples
LOAN_NBR	key				Remembered: you confirmed this as loan or application number on 2026-09-25	0% L0000000, L0000001, L0000002
FICO	fico		620, 680, 740		Remembered: you confirmed this as a FICO score on 2026-09-25	0% -9999, 680, 708
CHANNEL	category				Remembered: you confirmed this as a category to cut by (channel on 2026-09-25)	0% Online, Branch, Broker
ORIG_BAL	booked				Remembered: you confirmed this as booked or loan amount, which weights the reporting-level rates on 2026-09-25	0% 7655.75, 8189.94, 35307.6
BAD_FLAG	outcome			1 value(s) are neither 0 nor 1. They're left out of the outcome rates and counted; if they mean something, fix the extract.	Remembered: you confirmed this as the yes/no outcome: charged off, ever delinquent, decided by the system on 2026-09-25	0% 0, 2, 1
GCO_AMT	gco			1 value(s) aren't numbers among 4,999 that are. They're left out of any rate using this column and counted.	Remembered: you confirmed this as gross charge-off dollars on 2026-09-25	0% 0.0, #N/A, 2899.4
RANR_AMT	ranr				Remembered: you confirmed this as RANR dollars on 2026-09-25	0% -143.52, -177.5, 139.27
CURR_STATUS	servicing				name contains 'status';	0% 60DPD, ChargedOff, Current
DTI	dti				name contains 'dti'; a ratio with median 30%	0% 0.479, 0.360, 0.429

Origination Cube

Where does the book bleed? Every decision is made in the workbook; this window runs it.

Extract: /tmp/walk2/data/Consumer book Q3.csv

Browse...

1. Set up from this extract

2. Run the cube

Open the workbook

Ran on 5,000 loans; 66 tie-out checks agree.
Worst for Outcome, share of loans: fico under 620 / channel Broker.
Worst for Outcome, share of booked dollars: fico under 620 / channel Broker.
Worst for GCO per booked dollar: fico under 620 / channel Broker.
Remembered 11 confirmed answers for next time.
Open Consumer book Q3 - Origination Cube.xlsx: the results are on Where it bleeds.

Wrong turn F: the workbook copied to another folder. The Extract box names a 3,000-loan extract, and the run says 5,000 loans. It ran the original folder's extract and wrote the results into the copied workbook:

Origination Cube

Where does the book bleed? Every decision is made in the workbook; this window runs it.

Extract:

Ran on 5,000 loans; 66 tie-out checks agree.

Worst for Outcome, share of loans: fico under 620 / channel Broker.

Worst for Outcome, share of booked dollars: fico under 620 / channel Broker.

Worst for GCO per booked dollar: fico under 620 / channel Broker.

Remembered 11 confirmed answers for next time.

Open Consumer book Q3 - Origination Cube.xlsx: the results are on Where it bleeds.

How this was walked

On a Linux test machine:

- **The window:** Origination Cube.pyw's window on a virtual display, with its buttons pressed through driver/drive.py, and each state photographed.
- **Excel's part:** done cell by cell with openpyxl (driver/edit.py), standing in for picking from the dropdowns and typing.
- **The workbook:** each state printed through LibreOffice and photographed page by page (driver/render.sh).
- **The build:** frozen at commit f2f1032 with git archive, because src/ was being edited during the walk.
- **Memory and preferences:** a scratch CUBE_MEMORY and HOME.

The order actually walked:

1. Steps 1 to 15.
2. Step 17 (Set up again).
3. Step 18 (Forget).
4. Wrong turns A and A2, then step 16 (own edges).
5. Wrong turns B, C and D.
6. Wrong turns E and F.